FINAL YEAR PROJECT / SEMINAR

AGRI VISION

Federated learning for crop-disease detection

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BANANA • CAULIFLOWER • CHILI • GROUNDNUT • RADISH

FIVE CROPS. THIRTY DETECTION CLASSES.



Banana



Cauliflower



Chili



Groundnut



Radish

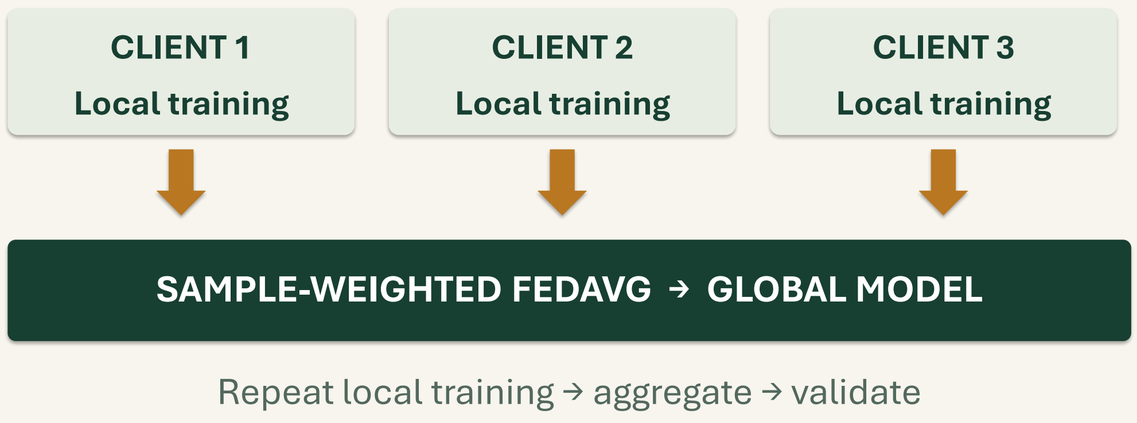
Examples from the labelled dataset; these photographs are not model predictions.

01 THE CHALLENGE

Visible disease symptoms can be small, similar across classes, and unevenly represented in crop datasets. Agri Vision explores automated symptom localisation while training a shared model across separate client partitions.

02 OUR APPROACH

YOLOv8n detects symptom regions and predicts their classes. Three clients train locally from a shared checkpoint; sample-weighted FedAvg combines their model weights.



Prototype: sequential three-client simulation on one Tesla T4. This is not a deployed multi-device system.

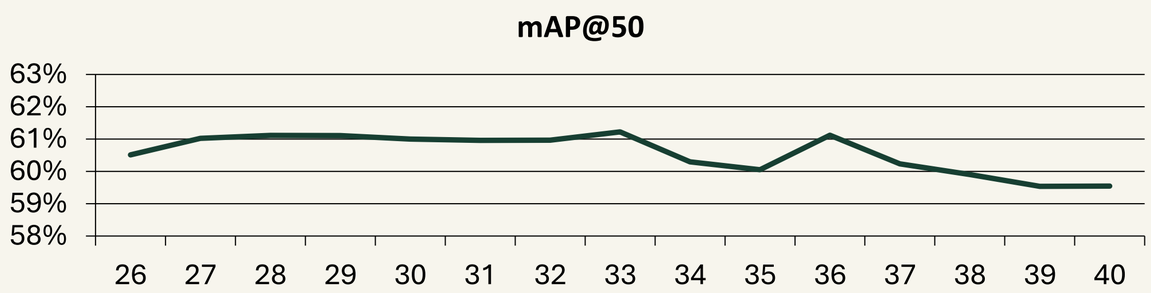
03 HELD-OUT TEST RESULTS

Round 35 • All 30 classes • 2,191 test images

63.72% Precision	**54.81%** Recall
55.72% mAP@50	**37.26%** mAP@50–95

Macro averages across classes. Precision and recall use the evaluation operating point selected over all 30 classes.

04 TRAINING TREND



Rounds 26–40 · Client-weighted validation mAP@50

WHAT THE RESULTS SHOW

The latest checkpoint was not the strongest.

Among rounds 26–40, validation mAP@50 peaked at 61.22% in round 33, then reached 59.55% in round 40. Validation mAP@50–95 peaked at 40.61% in round 30. These validation scores are separate from the held-out test results above.

NEXT: review weak-class labels and data balance; select checkpoints on validation performance; run a controlled Faster R-CNN comparison.

DATA SPLIT 15,512 train / 4,374 validation / 2,191 test • MODEL YOLOv8n + FedAvg